

Experiment: Prolog Program for DOB Database

1. AIM

To write and execute a **Prolog program to create a database named DOB** for storing and retrieving the **date of birth of persons** using facts, rules, and queries.

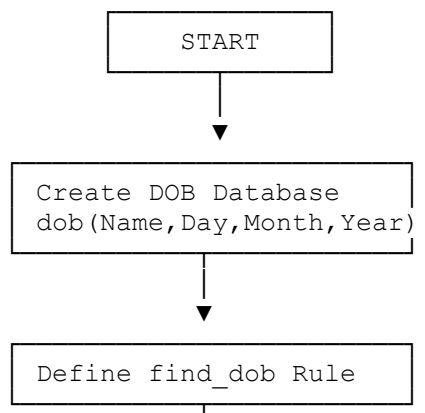
2. OBJECTIVE

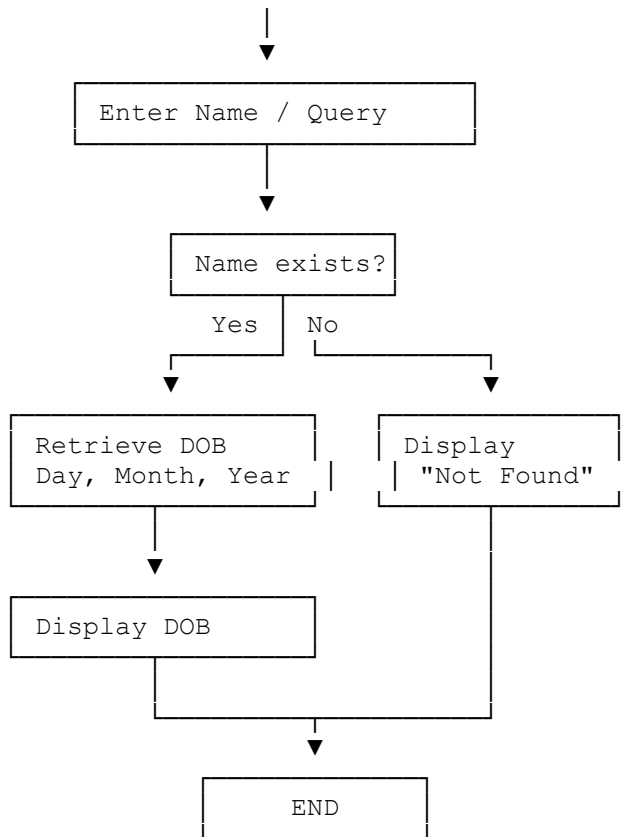
- To understand the concept of **database representation in Prolog**.
- To learn how to represent information using **facts**.
- To retrieve information using **rules and queries**.
- To create a database containing **name, day, month, and year of birth**.
- To understand Prolog's pattern matching and logical search.

3. ALGORITHM

1. Start the program.
2. Create a Prolog database using the predicate dob/4.
3. Store the person's **name, day, month, and year** as facts.
4. Define a rule find_dob/4 to retrieve the date of birth.
5. Enter the name of the person as a query.
6. Search the dob database for the given name.
7. If the name exists, retrieve and display the person's date of birth.
8. If the name does not exist, Prolog returns false.
9. Stop the program.

4. FLOWCHART:





Code with output:

The screenshot shows the SWISH Prolog IDE interface. On the left, the code editor contains the following Prolog code:

```

1 % DOB Database
2
3 dob(chandra, 17, 1, 2005).
4 dob(ravi, 10, 5, 2004).
5 dob(rahul, 25, 12, 2003).
6 dob(priya, 8, 8, 2005).
7 dob(anil, 15, 3, 2004).
8
9 % Rule to find date of birth
10
11 find_dob(Name, Day, Month, Year) :-
12     dob(Name, Day, Month, Year).
  
```

On the right, the output window shows the result of the query `find_dob(chandra, D, M, Y).`:

```

D = 17,
M = 1,
Y = 2005
  
```

Below the output, the query `?- find_dob(chandra, D, M, Y).` is shown in a text area. At the bottom right, there are buttons for "Examples", "History", "Solutions", "table results", and "Run".

RESULT

The Prolog program was successfully implemented to create a **DOB database** and retrieve the date of birth of a person using Prolog facts, rules, and queries.

For example:

?- find_dob(chandra, D, M, Y).

D = 17,

M = 1,

Y = 2005.

CONCLUSION

Thus, the **DOB database was successfully implemented using Prolog**. The experiment demonstrates how Prolog can represent data using facts and retrieve information using rules and logical queries.